CanonGate

Autonomous Pre-Publication Brand & Regulatory Governance

The Zero-Hallucination, Mathematically Bounded Gatekeeper for Regulated Enterprise Marketing

Formal SMT Prover
Z3 symbolic numeric & interval boundary verification

Conformal Risk Gate
Mathematically bounded false-positive miscoverage rate

RFC 9162 Merkle Tree
Immutable cryptographic audit trail & C2PA credentials

Tech Circuit Hackathon 2026 • Enterprise AI & Governance Track • Open Source Production Release

CanonGate • Confidential

The $280B Cost of Post-Publication Non-Compliance

The Compliance Explosion

- Global Regulatory Crackdown: RBI Fair Lending Practices, ASCI Disclosures, SEBI Investment Ad Codes, US FTC Endorsement Guides, and EU AI Act Article 52.
- Disproportionate Financial Fines: Regulatory penalties exceed ₹10 Cr+ ($1.2M+) per infraction, plus mandatory public retractions and class-action risk.
- The GenAI Velocity Paradox: Marketing teams generate 100x more campaign variants across social, email, web, and banners than human legal teams can audit.
- Irreversible Harm: Once non-compliant marketing collateral is published to CMS or social feeds, the regulatory violation is permanent and legally actionable.

By The Numbers

$280B+
Annual Global Corporate Compliance & Penalty Exposure

4.2 Days
Average Legal Turnaround Time per Campaign Asset

78%
Of Violations Occur in Fine Print, Disclosures & Images

CanonGate • Confidential

Why Traditional Compliance Solutions Fail

1. Manual Legal Review

Slow, Expensive & Inconsistent
- 3–7 day review bottlenecks stall marketing cycles
- High legal billable hours ($500+/hr)
- Inconsistent human subjective interpretation
- Fails completely under real-time automated CMS pipelines

2. Post-Hoc Scanners / DAMs

Too Little, Too Late
- Audits content after publication to staging or prod
- Zero pre-flight blocking on CMS webhooks
- Isolated from creator editing environments (Cursor, Figma)
- No formal mathematical bounds on claim veracity

3. Naive LLM Guardrails

Stochastic & Hallucination-Prone
- Non-deterministic: same ad yields different decisions
- Zero mathematical false-positive guarantees
- No courtroom-defensible cryptographic proofs
- Vulnerable to jailbreaks and prompt injections

CanonGate: The Autonomous Pre-Flight Gatekeeper

CanonGate shifts compliance from reactive post-publication damage control to an autonomous, deterministic pre-publish gate that intercepts and formally verifies content before it goes live.

Core Innovations

- Tri-Tier Formal Engine: Combines deterministic AST checks, Z3 SMT constraint solving, and SOTA 6-gate surgical rewriter.
- Conformal Risk Control: Mathematically bounds false-positive miscoverage rate ($\alpha \in [0.01, 0.25]$).
- RFC 9162 Merkle Tree Ledger: Tamper-evident, hash-chained audit log with $O(\log N)$ inclusion proofs.
- Multimodal Vision: OpenCV WCAG contrast, Tesseract OCR font height analysis, and CIEDE2000 color auditing.

Developer & Enterprise Integrations

- Pre-Publish Webhook Gate (**`/api/gate/check`**): Sub-18ms audit for Shopify, Webflow, and CI/CD pipelines.
- Model Context Protocol (MCP 2026-07-28): Real-time audit & fix in Cursor, Claude Desktop, and Antigravity.
- Air-Gapped DPDP 2025 Local Scrubber: Masks PAN, GSTIN, Aadhaar, IFSC, UPI locally before optional cloud calls.
- Enterprise Studio: 3 distinct personas for Creators, Brand Managers, and Legal Counsel.

CanonGate • Confidential

Tri-Tier Verification & Formal SMT Proving

Tier 1: Deterministic AST

Ultra-Fast Static Engine (< 5ms)
• Text & Headings: Banned terminology, forbidden claims
• Vision Analysis: WCAG 4.5:1 text-to-background contrast
• Typography: Font height bounding box ratios
• Table Grid: Alignment & column structure audits

Tier 2: Z3 SMT & Slot NLI

Mathematical Constraint Prover (< 25ms)
• Z3 Theorem Prover: Symbolic interval arithmetic over numeric rates, APR caps, and loan tenures
• MiniCheck Slot NLI: Sub-claim fact decomposition and entailment checking
• Contradiction Detection: Identifies subtle legal inconsistencies

Tier 3: SOTA 6-Gate Rewriter

Constrained Surgical Synthesis
• DPDP PII Scrubber: Local tokenization
• 6 Quality Gates: Length ratio, PII integrity, numeric anchor lock, edit distance <= 35%, re-verification, zero collateral
• Gemini 3.x Cascade: Multi-model fallback
• Circuit Breaker: < 1ms failover to AST

CanonGate • Confidential

Conformal Risk Control: Bounded False-Positive Guarantees

Why Static Thresholds Fail

Traditional compliance tools rely on static heuristics (e.g. `severity > 70`). In high-stakes regulatory domains, arbitrary cutoffs create catastrophic failure modes:
• Over-blocking: Rejects creative marketing copy, destroying conversion rates.
• Under-blocking: Allows regulatory non-compliance through subtle edge cases.

The CanonGate Conformal Solution:
Using Split-Conformal Prediction, CanonGate maps calibration non-conformity scores to a mathematically bounded miscoverage parameter $\alpha \in [0.01, 0.25]$:

$$\mathbb{P}(\text{Loss}(\lambda^*) > \varepsilon) \leq \alpha$$

This guarantees compliance officers an exact, mathematically proven risk ceiling.

Two-Tier Verdict Structure

🚫 BLOCKER (Severity >= 50)

Hard statutory violations: missing RBI KFS, prohibited medical claims, illegible contrast (< 3:1). Pre-publish gate immediately halts deployment.

⚠ WARNING (Severity < 50)

Stylistic deviations, secondary brand colors, non-critical fine print. Requires reviewer sign-off with recorded audit friction.

CanonGate • Confidential

Courtroom-Defensible RFC 9162 Merkle Tree Ledger

Cryptographic Immutability

• RFC 9162 Merkle Accumulator: Every submission, audit finding, reviewer action, and surgical rewrite is committed into a cryptographic binary tree.
• Logarithmic Proofs: Generates compact $O(\log N)$ inclusion proofs and append-only consistency proofs.
• SHA-256 Hash Chaining: Workflow state transitions are bound by strict hash chaining: $H_i = \text{SHA-256}(H_{i-1} \parallel E_i)$.
• Zero Tamper Window: Internal databases cannot alter historical records without breaking the Merkle root.

Offline Verifiability & C2PA Manifests

Offline CLI Verifier (`canongate verify`)

Regulators, legal counsel, and auditors can independently verify compliance certificates offline without database access:

```
$ canongate verify audit_certificate.json
[OK] Merkle root valid. Hash-chain verified across 18 events.
```

C2PA Content Credentials Embedding

• Approved digital assets are cryptographically signed with CanonGate's governance credentials.
• Ad networks and consumers can inspect origin and compliance proofs directly in browser.

Omnichannel CMS Webhooks & Model Context Protocol (MCP)

Pre-Publish Webhook Gate (`/api/gate/check`)

Synchronous Pre-Flight Deployment Blocker
• Sub-18ms Latency: Audits copy, tables, and images in milliseconds.
• Shopify Adapter: Automatically parses product titles, HTML bodies, variant pricing tables, and hero image contrast.
• Webflow Adapter: Ingests CMS rich-text fields, summary copy, and promotional banners.
• CI/CD Pipelines: Integrates into GitHub Actions to block non-compliant PRs before merge.

Model Context Protocol (MCP 2026-07-28)

IDE & AI Coding Assistant Native
• Stdio CLI: Run `canongate mcp` in Claude Desktop, Cursor, and Antigravity.
• 6 Native Tools: `verify_campaign`, `list_canons`, `inspect_canon`, `propose_fix`, `explain_rule`, `batch_evaluate`.
• 120-Asset Batching: Audits entire backlogs in seconds with fault isolation.
• Streamable HTTP: `POST /api/mcp` with DNS rebinding security.

CanonGate • Confidential

Privacy-by-Design & Indian DPDP 2025 Compliance

Air-Gapped Local Operation by Default

• Zero Mandatory API Keys: Core rule parsing, Z3 SMT solving, visual contrast checks, Merkle logging, and deterministic AST rewriting run 100% offline.
• Zero Customer Data Exfiltration: Customer campaigns, financial spreadsheets, and proprietary collateral never leave the local container by default.
• Transparent Disclosure: The UI and API explicitly disclose when running in air-gapped deterministic mode, preventing false claims of cloud AI generation.

Local DPDP 2025 PII Scrubber

Before any optional cloud LLM call (Gemini / OpenAI / Inkloom), text is scrubbed locally using Microsoft Presidio + Indian statutory regexes: • `PAN_ID` & `GSTIN` Tax Identifiers
• `AADHAAR_ID` 12-Digit UIDAI Tokens
• `UPI_VPA` & `IFSC` Banking Codes
• `PHONE` (+91) & `EMAIL` Addresses
• Bijective Verification: Validates placeholders before restoring locally.

CanonGate • Confidential

Market Opportunity & Target High-Stakes Verticals

FinTech & Lending

Digital Lending & Cards
- RBI Fair Practices Code
- APR & Fee Disclosures
- Guaranteed Approval bans
- Loan tenure minimums

D2C Nutra & Food

Supplements & Beauty
- ASCI Advertising Standards
- FSSAI / FDA efficacy rules
- “Miracle cure” bans
- Clinical claim contracts

Healthcare / Pharma

Telehealth & Diagnostics
- Prescription ad restrictions
- Diagnostic accuracy claims
- Statutory medical caveats
- Dosage contraindications

Enterprise Brands

Global Omnichannel
- Multi-brand design canons
- WCAG 2.1 AA accessibility
- Color contrast conformance
- FTC greenwashing audits

Total Addressable Market (TAM)

$42.8B
Global RegTech & Governance Market (2026)

21.4% CAGR
Annual Market Growth Rate

$8.5B
Serviceable Addressable Market (SAM)

CanonGate • Confidential

Enterprise SaaS Business Model & Unit Economics

Developer / Starter

Free / Open-Core
- CLI & Local MCP Stdio server
- Standard community Canons
- Offline SQLite audit storage
- Unlimited local AST audits

Team / Growth

₹39,999 / mo ($499/mo)
- Cloud Hosted REST API
- Shopify & Webflow Webhook Gates
- 50,000 automated checks / mo
- Conformal Risk Control slider
- 3-Persona Studio Collaboration

Enterprise Governance

₹1,99,999+ / mo ($2,499+/mo)
- Air-gapped on-premise container
- Custom Canon compiler pipeline
- RFC 9162 Merkle proof export
- Dedicated enterprise SLA (99.9%)
- SSO / SAML & role-based access

Exceptional Unit Economics

- Gross Margins > 85%: Deterministic Tier 1 & Z3 SMT checks cost under $0.0001 per check in compute.
- Zero API Cost by Default: Local execution eliminates dependency on expensive third-party LLM API tokens.
- High Expansion Revenue (NRR > 135%): check volume grows automatically with marketing content velocity.

Defensible Moats & The CanonGate Vision

Capability	Grammarly / Copy.ai	Traditional DAM	Generic LLMs	🛡️ CanonGate
Pre-Publish Deployment Gate	❌ Post-facto	❌ Staging only	❌ None	✅ Sub-18ms Webhook
Formal SMT Constraint Proving	❌ None	❌ None	❌ None	✅ Z3 SMT Solver
Mathematical Risk Bounds	❌ None	❌ None	❌ None	✅ Conformal α-Gate
Cryptographic Audit Proofs	❌ Plain logs	❌ DB rows	❌ None	✅ RFC 9162 Merkle
Model Context Protocol (MCP)	❌ None	❌ None	⚠ Basic	✅ Native 2026-07-28
Air-Gapped DPDP 2025 Local	❌ Cloud only	⚠ Partial	❌ Cloud only	✅ 100% Offline

The Operating System for Pre-Publication Digital Trust

Transforming how regulated enterprises create, verify, and publish content at scale.